

CS 543 — WEB SERVICES

ASSIGNMENT 2

Sweet Scoop — Services, Contracts & Schema

1. Capabilities

No.	Capability	Description
1	Register and authenticate customers	Create account, log in and access authorized features.
2	Browse ice creams and cakes	View products, categories, prices, flavours, sizes and details.
3	Search and filter products	Find products by name, category, flavour, price, size and availability.
4	Manage shopping carts	Add, update, remove and review items before ordering.
5	Place orders	Create an order from selected products, quantities and delivery information.
6	Process payments	Send payment requests and record payment results.
7	Track order status	View current status and progress of orders.
8	Manage products and availability	Administer products and availability.
9	Submit and manage reviews	Rate/review products and manage submitted reviews.
10	Manage customer profiles	Update profile information and delivery addresses.
11	Apply discounts and offers	Apply eligible coupons/promotional offers.
12	View order history	View previous orders, items, amounts and status history.

3. Service Contracts

Catalogue Service

Operation	Input	Output	Errors
listProducts	filters, limit, cursor	products + next link	400 invalid filters; 429 rate limit
getProduct	product reference	product details	404 product not found; 304 unchanged when applicable
checkAvailability	product reference, quantity	availability result	404 product not found; 422 invalid quantity; 503 dependency unavailable

Customer Service

Operation	Input	Output	Errors
getCustomer	customer reference	customer profile	404 customer not found; 401 unauthorized
createCustomer	customer details	customer confirmation	400 invalid input; 409 duplicate account
validateAddress	customer reference, address	validation result	400 invalid address; 404 customer not found

Order Service

Operation	Input	Output	Errors
placeOrder	customer, items, address, payment	order confirmation	400/401/404/409/422/502/503
getOrder	order reference	order details/status	401 unauthorized; 404 order not found
cancelOrder	order reference	updated order	401 unauthorized; 404 not found; 409 cannot cancel

Payment Service

Operation	Input	Output	Errors
createPayment	order/payment details	payment confirmation	400 invalid request; 409 duplicate; 502/503 provider failure
getPayment	payment reference	payment status	404 payment not found
refundPayment	payment reference	refund result	404 not found; 409 not refundable; 502 provider failure

Delivery Service

Operation	Input	Output	Errors
createDelivery	order/delivery details	delivery confirmation	400 invalid request; 409 duplicate; 503 delivery dependency failure
getDelivery	delivery reference	delivery/tracking status	404 delivery not found
getShippingQuote	delivery destination/details	shipping quote	400 invalid destination; 502/503 SOAP/dependency failure

4. Central Operation — placeOrder

placeOrder is the central operation of the Sweet Scoop ordering process.

Operation Specification

Item	Specification
Operation	placeOrder
Endpoint	POST /api/v1/orders
Authentication	JWT
Request	Idempotency-Key
Input	customer, items, address, payment
Validation	AJV validation
Success	201 Created — order confirmation
Error cases	400 Bad Request; 401 Unauthorized; 404 Not Found; 409 Conflict; 422 Unprocessable Entity; 502 Bad Gateway; 503 Service Unavailable

Processing Flow

1. Verify user → **2.** Validate cart → **3.** Check Catalogue availability → **4.** Calculate total → **5.** Process payment → **6.** Create delivery → **7.** Confirm order → **8.** Return response.

Internal Details Hidden from Callers

The operation hides internal database structures, SQL queries, identifier formats, inventory implementation, payment-provider implementation, delivery/SOAP implementation, and internal service orchestration. Callers interact only with the operation contract.

Reliability and Failure Handling

Outbound service calls use timeouts. The implementation plan specifies retry with increasing backoff and a circuit breaker for service dependencies. The Idempotency-Key supports safe handling of repeated logical requests.

6. Service Property Validation

Service	Reachable	Self-Contained	Has a Contract	Independent	Loosely Coupled
Catalogue	Yes — service API	Yes — owns catalogue data	Yes — operations defined	Yes — separate data ownership	Yes — API boundary
Customer	Yes — service API	Yes — owns customer/address data	Yes — operations defined	Yes — separate data ownership	Yes — API boundary
Order	Yes — service API	Yes — owns order data	Yes — placeOrder/getOrder/cancelOrder	Yes — separate data ownership	Yes — API-level dependencies
Payment	Yes — service API	Yes — owns payment data	Yes — operations defined	Yes — separate data ownership	Yes — provider hidden behind API
Delivery	Yes — service API	Yes — owns delivery/tracking	Yes — operations defined	Yes — separate data ownership	Yes — SOAP hidden behind boundary

Service Property Issues and Fixes

Issue	Property	Fix
Direct database access between services	Independent / Loosely Coupled	Use the owning service's API instead of accessing its database.
Long synchronous dependency chain in placeOrder	Loosely Coupled	Use narrow contracts, timeouts, retries with backoff and circuit breaker protection.
Database structures exposed as service interfaces	Has a Contract / Self-Contained	Expose operation-level contracts and hide implementation details.